Graph-Driven Simultaneous and Proportional Estimation of Wrist Angle and Grasp Force via High-Density EMG

Dongxuan Li, Peiqi Kang, Yang Yu, *Member, IEEE*, and Peter B. Shull

***Abstract*—Myoelectric prostheses are generally unable to accurately control the position and force simultaneously, prohibiting natural and intuitive human-machine interaction. This issue is attributed to the limitations of myoelectric interfaces in effectively decoding multi-degree-of-freedom (multi-DoF) kinematic and kinetic information. We thus propose a novel multi-task, spatial-temporal model driven by graphical high-density electromyography (HD-EMG) for simultaneous and proportional control of wrist angle and grasp force. Twelve subjects were recruited to perform three multi-DoF movements, including wrist pronation/supination, wrist flexion/extension, and wrist abduction/adduction while varying grasp force. Experimental results demonstrated that the proposed model outperformed five baseline models, with the normalized root mean square error of 13.2% and 9.7% and the correlation coefficient of 89.6% and 91.9% for wrist angle and grasp force estimation, respectively. In addition, the proposed model still maintained comparable accuracy even with a significant reduction in the number of HD-EMG electrodes. To the best of our knowledge, this is the first study to achieve simultaneous and proportional wrist angle and grasp force control via HD-EMG and has the potential to empower prostheses users to perform a broader range of tasks with greater precision and control, ultimately enhancing their independence and quality of life.**

***Index Terms*—High-density electromyography (HD-EMG), simultaneous and proportional control (SPC), spatial-temporal model.**

I. Introduction

HANDS play an indispensable role in daily activities, including interacting with machines [1], manipulating various objects [2], and communicating with people [3]. However, it is estimated that by 2050, more than 3.6 million individuals worldwide will undergo amputation [4], profoundly impacting their quality of life. Advanced prosthetic hands hold great promise in assisting amputees with hand-related activities and mitigating the difficulties they encounter [5]. Surface electromyography (sEMG) has emerged as the prevailing interface for prosthetic hands due to its noninvasiveness, portability, and wearability [6]. Nevertheless, the predominant myoelectric interfaces still remain limited in their ability to fully activate the dexterous functions of prosthetic hands, primarily due to the difficulty of decoding complex motor intentions into precise control commands, thus resulting in awkward and cumbersome human-machine interactions [7], [8]. Therefore, achieving intuitive and natural myoelectric control of prosthetic hands has proven to be a formidable challenge thus far.

Simultaneous and proportional control (SPC) has arisen as a promising approach for continuous myoelectric control, which allows for deciphering multi-degree-of-freedom (multi-DoF) motor intentions simultaneously and proportionately [9], replacing discrete pattern recognition (PR) control and the conventional dual-site direct control method [10]. Previous studies have achieved estimating hand and wrist joint angles with varying degrees of success [11], [12], [13], [14], [15]. In addition, investigations into deciphering multi-DoF kinetic intentions (force/torque) have explored semi-unsupervised [16], physics-based [17], [18], and deep learning models [19], [20]. Notably, the simultaneous decoding of kinematic and kinetic intentions for amputees can facilitate precise positioning and force control of prosthetic limbs. For example, the implementation of SPC in wrist kinematics and hand kinetics has the potential to empower amputees to grasp objects independently and adaptively with varying weights and positions by adjusting wrist orientation and grasp force, thereby enhancing the natural and intuitive execution of daily activities. However, research on SPC has predominantly been performed on either kinematics (joint angles) or kinetics (force/torque) in isolation. A primary challenge in realizing the SPC of multi-DoF kinematics and kinetics lies in the non-negligible crosstalk and intrinsic nonlinear characteristics of sEMG signals resulting from muscle synergy and common drive during multi-DoF movements [21], posing a substantial obstacle to effectively extract representative features and map them to independent kinematic and kinetic parameters.

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Deep learning methods have demonstrated the ability to automatically learn features in simultaneous and proportional control [22]. Among these methods, Long Short-Term Memory (LSTM) has been extensively utilized to capture temporal dependencies in sEMG signals [23]. In addition, previous studies have indicated that the spatial distribution patterns of sEMG signals can be influenced by muscle contractions and motor unit recruitment [24]. Therefore, introducing the spatial distribution characteristics of sEMG signals has the potential to further enhance the accuracy of SPC [25], [26]. Notably, high-density electromyography (HD-EMG) presents the advantage of recording finer muscular activations from multiple functional muscles, thus offering valuable spatial information [27]. Graph Convolutional Neural Network (GCN) excels in extracting spatial features by leveraging the inherent topological structure of graphs. Although several studies have reported the feasibility of GCN in PR for myoelectric control [28], [29], [30], its effectiveness in capturing spatial features for SPC remains unknown. Additionally, the prevailing deep learning models are limited to extracting features in a single domain. Introducing and integrating information from multiple domains of sEMG signals potentially enhances the performance of continuous motion decoding.

To address the above questions, this study aims to accomplish simultaneous and proportional multi-DoF kinematic and kinetic estimation via integrating spatial and temporal domain information derived from the HD-EMG signals. More specifically, the investigation concentrates on wrist pronation/supination, flexion/extension, abduction/adduction, and hand grasp force, which are crucial for proficiently governing the position and force execution of prosthetic hands.

The contributions of this paper are as follows:

1) We achieved simultaneous and proportional wrist angle and grasp force control via HD-EMG for the first time, offering a promising solution to enhance the naturalness of both position and force control in prosthetic hands.

2) We proposed an integrated model that combines GCN and LSTM to improve simultaneous wrist angle and grasp force estimation performance by leveraging the capability to automatically capture spatial-temporal features from HD-EMG.

II. Methods and Materials

A. Subjects

Twelve subjects (age: 23.5 ± 1.3 years; height: 178.8 ± 3.3 cm; weight: 73.1 ± 10.3 kg) voluntarily participated in this experiment. No previous history of neuromuscular or joint diseases was reported in the subjects. All subjects provided informed consent before participation. This experiment was performed in accordance with the Helsinki Declaration and was approved by the ethics committee of Shanghai Jiao Tong University (No. E2021013P).

B. Experimental Protocol

Real-time wrist joint orientations were captured by a commercial inertial measurement unit (IMU, Xsens-MTw, Xsens Technologies B.V., Netherlands), which was attached to the back of the hand and sampled the signals at a rate of 100 Hz (Fig. 1(a)). In addition, a grasp dynamometer (G100, Biometrics Ltd., U.K.) was deployed for grasp force measurement, with a sampling rate of 200 Hz and a rated load of 90 Kg (Fig. 1(a)). A customized software built in Matlab received real-time wrist joint orientation data, grasp force data, and further converted the orientation data to physical wrist angles [31]. The software also synchronized the acquisition of wrist angle and grasp force signals and provided visual feedback for the subjects.

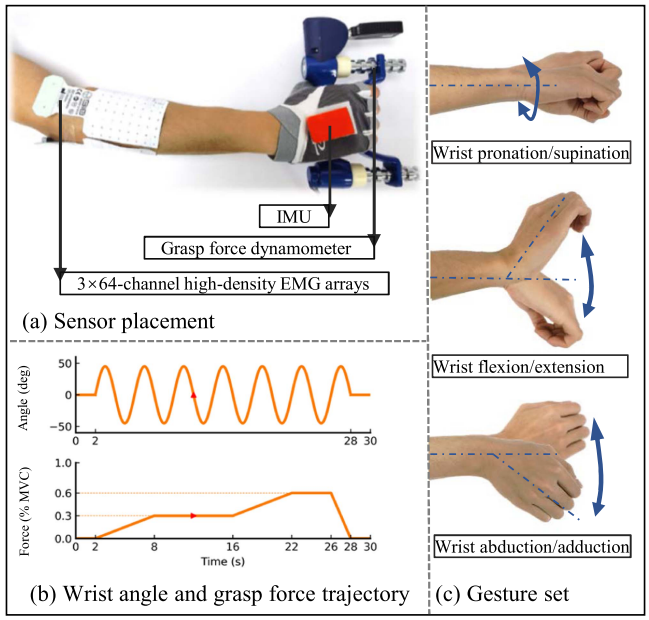

Fig. 1. Experimental paradigm. (a) Sensor placement: Three arrays of high-density electromyography (HD-EMG) electrodes were strategically positioned around the forearm muscles to record muscle activity. A grasp dynamometer was employed to measure grasp force. An inertial measurement unit (IMU) was affixed to the hand's back, capturing real-time wrist joint orientations. (b) Predefined wrist angle and grasp force trajectory: Wrist joint motion followed a sinusoidal curve. Grasp force exhibited a stepwise trajectory with two distinct levels (30% and 60% maximum volunteer contraction (MVC)). (c) Gesture set: wrist pronation/supination, wrist flexion/extension, and wrist abduction/adduction.

For HD-EMG signals, three arrays of 8 × 8 electrodes (GR10MM0808, OT Bioelettronica, Italy) with 10 mm inter-electrode distance were positioned to encircle and capture sEMG signals from the forearm muscles (Fig. 1(a)). Specifically, Electrode array 1 covered the extensor digitorum, extensor indicis, extensor carpi radialis longus, abductor pollicis longus, and extensor digiti minimi. Electrode array 2 was positioned to capture signals from the flexor carpi radialis, flexor digitorum superficialis, and pronator teres. Electrode array 3 was situated to detect activity from the extensor carpi ulnaris, flexor carpi ulnaris, and flexor digitorum profundus. These muscles were specifically targeted for controlling wrist movements and hand grasp, and their anatomical locations were determined through palpation. Subsequently, 192-channel monopolar sEMG signals were amplified with a gain of 150 and sampled at a rate of 2048 Hz using an amplifier (Quattrocento, OT Bioelettronica, Italy). Prior to the acquisition, we cleaned the skin with alcohol wipes and used conductive gel between electrodes and skin to

ensure signal quality. In addition, the sEMG, IMU, and force signals were simultaneously triggered by an Arduino development board.

The experimental paradigm consisted of three multi-DoF movements (Fig. 1(c)). Each movement comprised one DoF wrist motion and an isometric grasping motion of the hand, simultaneously activating two DoFs. Specifically, Movement 1 involved wrist pronation/supination and isometric grasping (Pro-Sup-Grasp). Movement 2 encompassed wrist flexion/extension and isometric grasping (Fle-Ext-Grasp). Movement 3 entailed wrist abduction/adduction and isometric grasping (Abd-Add-Grasp). The subjects were instructed to perform wrist movements following a predefined sinusoidal trajectory. Simultaneously, the grasping force exhibited a stepwise trajectory, containing two distinct force levels (30% and 60% maximum volunteer contraction (MVC)) (Fig. 1(b)).

During data collection, subjects were instructed to naturally sit on a chair, grasp the force dynamometer with their dominant hands, and maintain a neutral upper limb posture [32]. Specifically, the subjects were seated with their shoulder adducted and neutrally rotated, elbow flexed at 90°, forearm in a neutral position, and the wrist between 0° and 30° dorsi flexion and between 0° and 15° of ulnar deviation. The MVC for grasping was measured by instructing each subject to exert maximum effort while grasping the dynamometer for 3 seconds. This procedure was repeated three times, with a rest period of 2 minutes between each measurement. The average value of the three tests was calculated and considered as the MVC measurement result for the grasping gesture.

Real-time wrist angle and grasp force values were displayed on a screen in front of the subjects to provide visual feedback, assisting them in controlling their wrist movements and matching grasp force to the predefined trajectories. Once the subjects familiarized themselves with the experimental paradigm, they were instructed to perform the three movements sequentially. Each movement lasted for 30 seconds and was repeated eight times. A 1-minute break was provided between each trial to prevent muscle fatigue. The dataset, collected in one session, comprises a total of 288 files (calculated as 12 subjects × 3 movements × 8 trials). Each file contains synchronized data for 192 channels of HD-EMG, wrist angle, and grasp force, recorded over a 30-second interval, and is saved in the '.mat' format. Subsequent offline analysis and data processing were done through MATLAB R2021b and Python 3.9.

C. Signal Preprocessing

A fourth-order Butterworth bandpass filter with a range of 20 to 450 Hz and a notching comb filter centered on 50 Hz and its harmonics were utilized for the preprocessing of the raw sEMG signals. This processing step effectively removed low-frequency movements, attenuated high-frequency noise, and mitigated interference from power-line frequencies, ensuring that only relevant muscle activity frequencies were retained while minimizing unwanted sources of noise and interference. For the grasp force and wrist angle signals, a fourth-order Butterworth low-pass filter with a cutoff frequency of 1 Hz was utilized to attenuate high-frequency components. The force signals were then normalized to a range of [0, 1], while the angle signals were linearly transformed to a range of [−1, 1]. Additionally, the force signals were downsampled to match the frequency of the wrist angle signals.

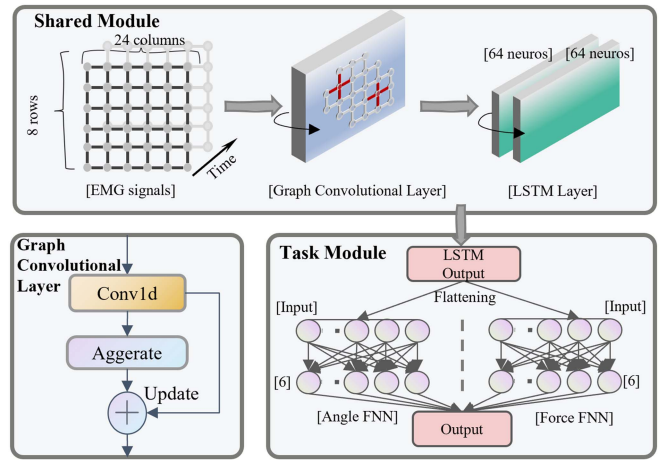

Fig. 2. Architecture of the proposed model (GCN-LSTM). In the Shared Module (upper part), high-density electromyography (HD-EMG) signals were transferred into graphical expressions. A Graph Convolutional Layer combined with two LSTM layers was utilized to capture spatial-temporal features for wrist angle and grasp force estimation. In the Task Module (lower right), two separate Fully Connected Networks (FCNs) were deployed to map the common spatial-temporal features to wrist angle and grasp force, respectively. A Graph Convolutional Layer (lower left) consists of three steps: a one-dimensional convolution operation, aggregating neighboring vertices, and updating target vertices.

The filtered sEMG signals were segmented using the sliding window method. The window size was empirically set at 100 ms without any overlap [33]. To effectively extract the information embedded in the sEMG time series, several time-domain features were calculated within each window, including the Mean Absolute Value (MAV), Waveform Length (WL), Standard Deviation (SD), and Root Mean Square (RMS) [34], [35]. A min-max normalization approach was then applied to each feature dimension.

D. GCN-LSTM Model

A multi-task model (GCN-LSTM) based on HD-EMG was proposed to estimate wrist angle and grasp force simultaneously (Fig. 2). The HD-EMG signals were spatially distributed in an 8 × 24 array (vertex number: 192, edge number: 161) in a regular Euclidean graph, where each vertex had a feature dimension of four. The adjacency matrix A (192 × 192) was computed for the HD-EMG graph, where one indicated connectivity between vertices, and zero meant disconnection. HD-EMG signals were then divided along the temporal dimension into segments of length six. In the Shared Module, the embedded HD-EMG graph tensor with the dimension of (6 timesteps × 192 vertices × 4 features) was fed into the graph convolutional layer. The spatial feature extraction process is shown as follows:

$$h_{output} = h_{input}w \oplus Ah_{input}w \tag{1}$$

Where h_{input} represents the input of the graph convolutional layer, h_{output} represents the output of the graph convolutional layer, and w is the weight of the one-dimensional convolutional layer (Conv1d).

The Conv1d (10*1@1) layer first operated on the spatial dimension of the input HD-EMG graph tensor, linearly mapping the low-dimensional features of the original input to a higher dimension. By leveraging the weight-sharing property of CNN, the time complexity of model training can be reduced. The adjacency matrix A aggregated the features of all first-order neighboring vertices of target vertices by summation. Subsequently, we employed a residual connection structure, updating target vertices by concatenating the features of target vertices with the aggregated features of their neighboring vertices. A two-layer LSTM network was then employed to extract temporal dependencies. Each LSTM layer comprised 64 neurons with the tanh activation function.

In the Task Module, a Flatten layer is deployed to flatten the output feature tensor from the LSTM layer. Two separate fully connected networks (FCNs) were set up for the wrist angle and grasp force estimation tasks. Both FCNs consisted of only one output layer, with six neurons, which was consistent with the length of the segmented EMG signals. The two FCNs could map the shared spatial-temporal feature extracted in the Shared Module to the wrist angle and grasp force, respectively.

An Adaptive Moment Estimation (Adam) optimizer with an initial learning rate equal to 0.001 was used to train the proposed model. The number of iterations and the batch size were set as 100 and 32, respectively. Mean Square Error (MSE) loss was deployed on angle and force estimations, respectively, due to the advantage of fast convergence. The final loss of output is calculated as follows:

$$LOSS = a_1 MSE_{angle} + a_2 MSE_{force} \tag{2}$$

where a_1 and a_2 were hyper-parameters used to balance the loss of angle and the loss of force and were both set at 0.5 through the pilot study.

The Reduce on Plateau strategy was implemented to adjust the learning rate automatically. When the loss on the validation set did not decrease for ten consecutive epochs, the learning rate was multiplied by a drop factor of 0.5. Additionally, the Early Stopping strategy and the Model Checkpoint strategy were employed during model training to mitigate overfitting [36], [37]. Specifically, the training process was structured to automatically terminate if the loss on the validation set did not decrease for 15 consecutive epochs. The model was configured to save its weights after each epoch only when its performance on the validation set surpassed the previous best performance. By consistently monitoring the model's performance on the validation set throughout the training process and halting the training when no further improvement is observed, we effectively safeguard against overfitting the model to the training data.

Leave-one-trial-out cross-validation was deployed to assess the performance of the proposed model in this study. The process involved training the model using data from six randomly chosen trials while designating one another trial as the validation set to tune hyper-parameters and select the optimal model based on minimizing the loss observed in the validation set. The selected model was subsequently evaluated using the remaining data from the excluded trial. This partitioning scheme was repeated iteratively until all trials had been utilized as a test set, ensuring a comprehensive and unbiased evaluation of the model's performance.

E. Baseline Models

We conducted a comprehensive performance comparison of our proposed model with two classical machine learning models, Support Vector Regression (SVR) and Gaussian Process Regression (GPR), as well as three deep learning models, LSTM, Convolutional LSTM (ConvLSTM) [38], and CNN. The parameters of these five baseline models were manually optimized.

For SVR, we deployed a radial basis kernel with a regularization parameter set to 10. For GPR, the mixed kernel functions were obtained by multiplying a constant kernel with a radial basis kernel having a length scale of 10. The original feature dimension of the input data for SVR and GPR was 768 (4 features $\times$ 192 channels). To enhance computational efficiency, Principal Component Analysis (PCA) was applied to reduce the dimensionality of the data while preserving over 98% of the variance.

In the case of LSTM, the network was optimized with two layers, each comprising 64 neurons. Following the LSTM layers, two FCNs with six neurons each were connected for angle and force estimation. The input data was preprocessed using PCA to reduce dimensionality and was then segmented along the time dimension into segments of length six. For ConvLSTM, the model was configured with 128 2-dimensional ConvLSTM kernels of size 4 $\times$ 4, 64 kernels of size 3 $\times$ 3, and 32 kernels of size 2 $\times$ 2, introducing the convolution operation in the gates of the LSTM layer to capture spatial-temporal features. A 3-dimensional convolutional layer and a Flatten layer were employed to reshape the data dimension. Additionally, two separate FCNs were connected for the angle and force estimation tasks. As for CNN, we adopted the model from [27], which demonstrated success in simultaneous and proportional multi-DoF wrist torque estimation based on HD-EMG. The performances of these models were evaluated using leave-one-trial-out cross-validation.

F. Data Analysis

The normalized root mean square error (nRMSE) and correlation coefficient (R) were utilized as evaluation metrics for estimation accuracy in this study. nRMSE is less sensitive to the scale of the variables, making it suitable for evaluating the performance of regression models with diverse input and output ranges. R could gauge the strength and direction of the linear relationship between variables, allowing for a comprehensive assessment of the association between estimation and ground truth. Lower nRMSE and higher R values indicate better estimation accuracy. The definitions of nRMSE and R are shown as

follows:

$$\text{nRMSE} = \frac{\sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}}{\max(y) - \min(y)} \tag{3}$$

$$R = \frac{\sum_{i=1}^{n}(\hat{y}_i - \bar{\hat{y}})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(\hat{y}_i - \bar{\hat{y}})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}} \tag{4}$$

where y_i and $\hat{y}_i$ represent recorded wrist angle/grasp force and predicted wrist angle/grasp force, respectively; n is the total number of angle/force series.

Before conducting the significance analysis, all results were assessed for normal distribution using the Shapiro-Wilk test, with a significance value of $p > 0.05$. We implemented the significant analysis with a two-tailed, paired t-test to check whether there were differences between wrist angle estimation accuracy and grasp force estimation accuracy within three multi-DoF movements. In addition, we employed ANOVA to assess the potential impact of regression methods on the accuracy of wrist angle and grasp force estimation. We also examined whether the number of electrodes significantly influenced estimation accuracy for wrist angle and grasp force. Before conducting the ANOVA analysis, we utilized Levene's test to verify the homogeneity of variance for the values of R and nRMSE. A significance value higher than 0.05 indicates the variance is homogeneous. If so, we conducted post hoc analysis using the Least Significant Difference (LSD) method. If not, Tamhane's T2 test was used instead. The statistical significance level for post hoc analysis was set at $p = 0.05$.

III. Results

A. Results of Different Methods

The proposed model exhibited superior performance compared with five baseline models, achieving the lowest average nRMSE of 13.2% for wrist angle estimation and 9.7% for grasp force estimation (Fig. 3). For wrist angle estimation, LSTM, ConvLSTM, SVR, GPR, and CNN demonstrated the average nRMSE of 19.5%, 19.5%, 16.8%, 22.5%, and 16.4%, respectively. Statistical analysis indicated significant differences between GCN-LSTM and the five baseline models ($p < 0.05$). For grasp force estimation, LSTM and SVR exhibited relatively lower nRMSE, with average nRMSE of 10.9% and 11.4%, respectively. It was shown that there were no significant differences between GCN-LSTM and LSTM ($p > 0.05$). However, ConvLSTM, GPR, and CNN performed worse than the proposed model. Statistical analysis revealed that the proposed model significantly outperformed SVR ($p < 0.05$), ConvLSTM ($p < 0.05$), GPR ($p < 0.05$), and CNN ($p < 0.05$). In addition, individual differences had a considerable impact on grasp force estimation accuracy ($p < 0.05$).

The average correlation coefficient (R) values across all subjects for wrist angle estimation were 89.6% for GCN-LSTM, 71.5% for LSTM, 58.2% for ConvLSTM, 80.7% for SVR, 67.2% for GPR, and 81.6% for CNN (Fig. 4). Significant differences were identified between GCN-LSTM and the other baselines ($p < 0.05$). For grasp force estimation, GCN-LSTM and LSTM achieved similar R values, reaching 91.9% and 88.7%, respectively, which were significantly higher than SVR, ConvLSTM, GPR, and CNN ($p < 0.05$). In addition, the statistical analysis revealed that individual differences had a significant impact on wrist angle estimation and force estimation ($p < 0.05$).

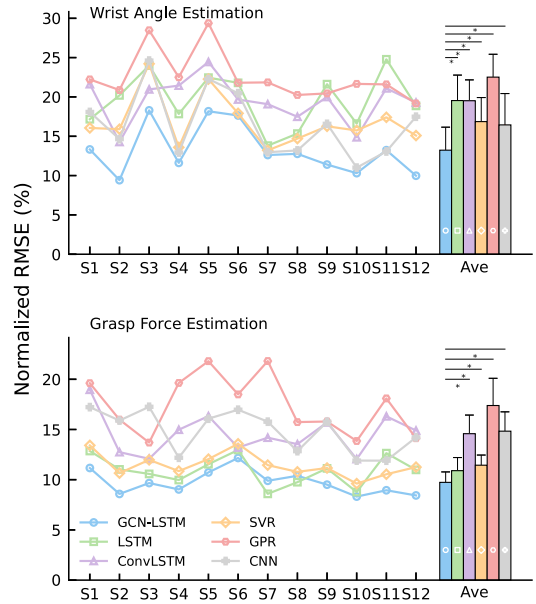

Fig. 3. nRMSE for wrist angle estimation and grasp force estimation. The performances of GCN-LSTM, LSTM, ConvLSTM, SVR, and GPR of each subject are presented. The histogram and error bar represent mean values and standard deviation across 12 subjects, respectively. * denotes a significant difference between the proposed model and other baselines.

B. Results of Wrist Angle and Grasp Force Estimation

The estimated wrist angles were found to be in good agreement with the recorded wrist angles within the three multi-DoF movements (Fig. 5). However, the proposed model faced challenges in predicting peak wrist angle values. Similarly, the estimations of grasp force also demonstrated good agreement with the ground truths within the three multi-DoF movements, although slight oscillation was observed (Fig. 5).

The comparative results of wrist angle estimation and grasp force estimation are presented in Fig. 6. The nRMSE values of grasp force estimation within the three multi-DoF movements were 9.6%, 10.1%, and 9.5%, respectively. The nRMSE values of wrist angle estimation were higher than those for grasp force estimation, with 14.2%, 13.0%, and 12.5% in Pro-Sup-Grasp, Fle-Ext-Grasp, and Abd-Add-Grasp, respectively. Significant differences were found between the nRMSE of grasp force estimation and that of wrist angle estimation in Pro-Sup-Grasp ($p < 0.01$), Fle-Ext-Grasp ($p < 0.05$), and Abd-Add-Grasp ($p < 0.05$). Regarding the R metric, grasp force estimation demonstrated high accuracy, achieving 92.8%, 90.6%, and 92.3% in Pro-Sup-Grasp, Fle-Ext-Grasp, and Abd-Add-Grasp, respectively. In addition, wrist angle estimation exhibited comparable performance to grasp force estimation. There were no significant differences

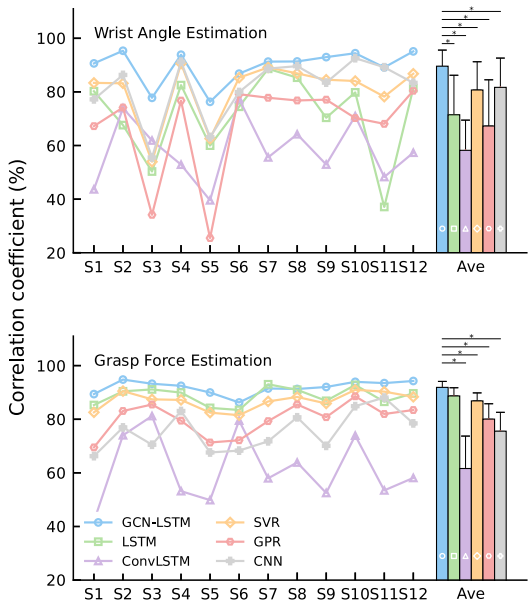

Fig. 4. Correlation coefficient (R) for wrist angle estimation and grasp force estimation. The performances of GCN-LSTM, LSTM, ConvLSTM, SVR, and GPR of each subject are presented. The histogram and error bar represent mean values and standard deviation across 12 subjects, respectively. * denotes a significant difference between the proposed model and other baselines.

observed between the R values of grasp force estimation and the R values of wrist angle estimation in each movement ($p > 0.05$).

C. Results of Reducing the Number of Electrodes

The relationship between estimation accuracy and the number of electrodes is depicted in Fig. 7. We selected 192, 96, 64, 48, 40, 32, and 24 electrodes from the three 8×8 electrode slices, with 24, 12, 8, 6, 5, 4, and 3 evenly spaced rows of electrodes along the forearm circumference, retaining the electrodes aligned with the direction of the muscle fibers.

Wrist angle estimation performed best when using all (192) electrodes, with the nRMSE of 13.2% and the R of 89.6%. As the number of electrodes reduced, the value of nRMSE increased slightly, rising at 14.9% when using 40 electrodes. In addition, the R value decreased accordingly with the number of electrodes reducing, which dropped to 86.3% when using 40 electrodes. Statistical analysis showed no significant differences when using 192, 96, 64, 48, and 40 electrodes ($p > 0.05$). However, as the number of electrodes further decreased to 32 and 24, the nRMSE increased to 15.8% and 16.5%, respectively, which were significantly higher than that of using all electrodes ($p < 0.01$). The R value also decreased to 84.9% when using 32 electrodes and reached its lowest value of 82.3% when using 24 electrodes. It was found that there were significant differences between using 24 electrodes and all (192) electrodes ($p < 0.01$).

Similarly, grasp force estimation accuracy was the highest when using all 192 electrodes, with the nRMSE of 9.7% and the R of 91.9%. The nRMSE only increased by 0.5%, and the R dropped by 1.1% as the number of electrodes decreased from 192 to 40. In addition, the estimation accuracy decreased slightly when reducing to 32 electrodes, with the nRMSE of 10.8% and the R of 89.5%. When using 24 electrodes, the nRMSE and R values were 10.8% and 89.7%, respectively. Statistical analysis demonstrated no significant differences when the number of electrodes decreased from 192 to 24 ($p > 0.05$).

IV. Discussion

This article proposed and validated the feasibility of a multi-task spatial-temporal model driven by graphical HD-EMG for simultaneous and proportional wrist angle and grasp force estimation. Experimental results showed that the proposed approach achieved satisfactory performance.

A. Analysis of the Estimation Results Within Three Movements

Previous research in the field of simultaneous and proportional control has primarily focused on the estimation of specific motor movements such as dexterous finger movements [39], wrist movements [40], grasp force [41], or wrist joint torque [27]. Only a few studies have explored the simultaneous decoding of both kinematic and kinetic motor intentions. Roy et al. [42] were the first to achieve concurrent and continuous prediction of finger force and joint angle based on motor unit decomposition, with an RMSE of 3.47% MVC and 5.40° for finger force and joint angle estimation, respectively. This study underscores the potential of utilizing motoneuron activities for predicting multi-DoF kinetics and kinematics, suggesting the possibility of incorporating such insights into data-driven methods to enhance estimation accuracy. Mao et al. [43] reported the highest correlation coefficient of 92.12% for wrist angle and grip force estimation based on seven sparse EMG electrodes. Despite the high correlation coefficient, this study was limited to exerting grip force under 30% MVC, which has a lesser influence on wrist angle estimation. This limitation might account for the high average estimation accuracy. From a physiological view, the exertion of force or the execution of joint movements often entails the synergy of the same muscle groups [21], intensifying the nonlinearity of sEMG signals.

Results of our study demonstrated that the estimation performance for grasp force was superior to that for wrist angle estimation in terms of nRMSE within the three multi-DoF movements, which aligned with previous physiological findings. Furthermore, our results demonstrated a substantial correlation (R values around 90%) between the proposed model's predictions and the actual wrist angle and grasp force values. This robust correlation underscores the model's efficacy in capturing linear relationships between input features and target outputs, highlighting its potential for accurate prediction of wrist movements and grasp forces. In addition, the accuracy of wrist angle prediction for the three DoFs (flexion/extension, abduction/adduction, and pronation/supination) was found to be comparable, without significant differences observed (Fig. 6). From an anatomical perspective, the prediction of pronation/supination is expected to exhibit significantly lower accuracy due to the involvement of deep-layer muscles, specifically pronator teres and pronator quadratus. These muscles generate weaker sEMG signals and

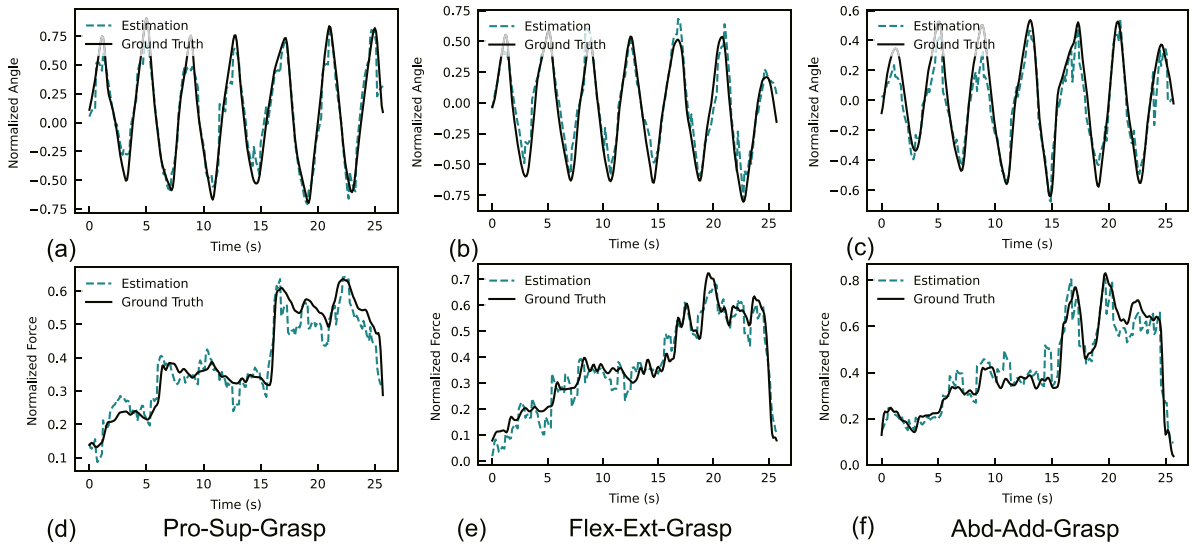

Fig. 5. Instances of subject 2 for the simultaneous and proportional estimation of wrist angle and grasp force. (a), (b) and (c) are wrist angle estimation results. (d), (e) and (f) are grasp force estimation results under three movements, respectively. Movement 1: wrist pronation/supination and isometric grasping (Pro-Sup-Grasp). Movement 2: wrist flexion/extension and isometric grasping (Fle-Ext-Grasp). Movement 3: wrist abduction/adduction and isometric grasping (Abd-Add-Grasp). The line in black denotes ground truth, while the dashed line in green denotes estimation. Normalized Force: The force values were normalized to a range of [0, 1] by dividing the maximum volunteer contraction (MVC). Normalized Angle: The angle values were linearly transformed to a range of [−1, 1].

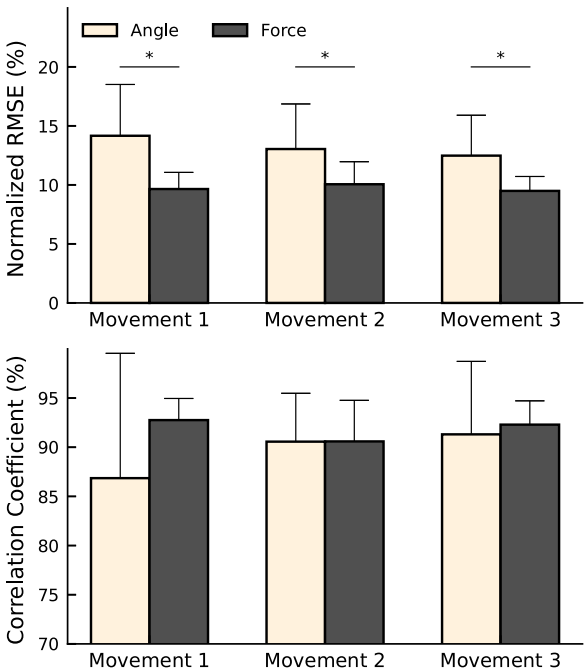

Fig. 6. Comparative results of wrist angle and grasp force estimation within three movements. The histogram and error bar represent mean values and standard deviation across 12 subjects, respectively. This figure indicates that grasp force estimation is more accurate than wrist angle estimation. * indicates statistical significance between the grasp force estimation accuracy and the wrist angle estimation accuracy.

are susceptible to crosstalk from superficial muscles, resulting in reduced accuracy [25]. The unexpected finding may be attributed to the use of HD-EMG, which mitigates the impact of muscle signal crosstalk from superficial muscles [44]. Additionally, these findings affirm the capability of the proposed model to extract deep-level features from HD-EMG signals affected by crosstalk.

B. Analysis of the Effectiveness of the Proposed Model

To validate the feasibility and effectiveness of the proposed GCN-LSTM model, we conducted a comparison with other commonly used methods in myoelectric continuous estimation, including LSTM, ConvLSTM, CNN, SVR, and GPR, which have been extensively studied in previous works [22], [23], [45]. Results in the presented study demonstrated that the proposed model achieved the highest estimation accuracy in SPC of wrist angle and grasp force compared to the other methods. The superiority observed can be attributed to the incorporation of the GCN module and the LSTM layer on HD-EMG. This combination captures representative spatial features and facilitates efficient learning of long-range dependencies in time-series data. Since the execution of movement and force involves intricate muscle synergy and the superposition effect of muscle fiber action potential [16], [42], exploiting the connections between synergized muscles has the potential to improve intention decoding accuracy. In this study, the utilized GCN module offers a distinctive advantage by mapping electrodes to nodes and representing the correlations between electrodes as edges in the graph representation. This unique approach facilitates the capture of correlation and topological information regarding muscle activation characteristics across various positions. Given that HD-EMG electrodes comprise numerous closely spaced channels, they simultaneously capture signals from different

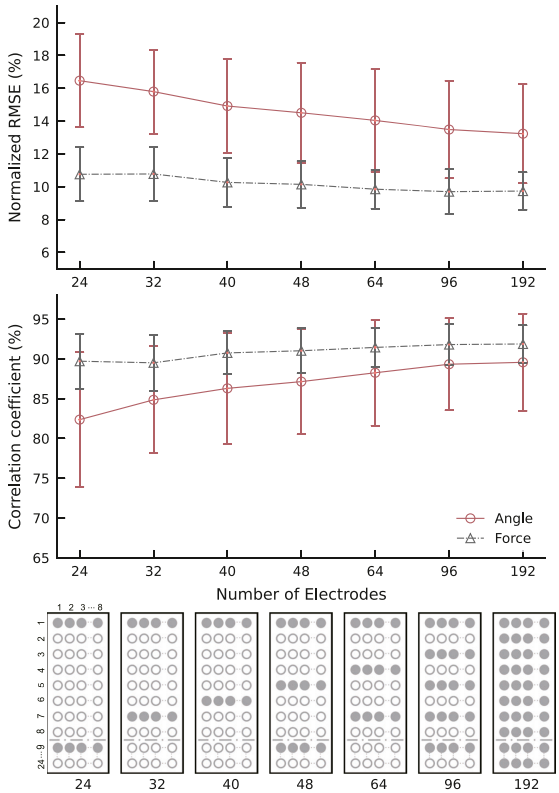

Fig. 7. Relationship between the estimation accuracy and the number of electrodes. nRMSE and correlation coefficient (R) for the different numbers of electrodes are presented, respectively. 24, 12, 8, 6, 5, 4, and 3 evenly spaced rows of electrodes along the forearm circumference were selected, ensuring that at least one row of electrodes was chosen for each electrode slice (a total of 3 slices). Electrodes aligned with the direction of the muscle fibers were retained. In total, 192, 96, 64, 48, 40, 32, and 24 electrodes were selected. A schematic diagram with 10 rows and 4 columns illustrating the electrode selection is presented. The black dots indicate the selected electrodes. The number of each row and column is denoted.

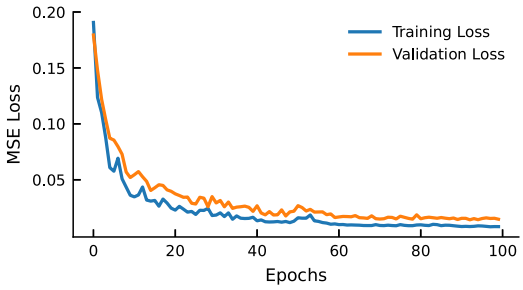

Fig. 8. Example of training loss and validation loss over epochs. The mean squared error (MSE) was employed as the loss criterion during the training process. The decreasing trends observed in both training loss and validation loss curves signify the convergence of the model's optimization progress over successive epochs.

parts of the muscle, providing intrinsic topological connections between electrodes placed on correlated muscles [28], [30].

It is noteworthy that LSTM exhibited comparable accuracy to the proposed model in grasp force estimation. This can be attributed to the highly nonlinear temporal patterns present in sEMG signals, which reflect muscle activation levels at different moments, making LSTM well-suited for capturing features in time-series data. However, the performance of LSTM in wrist angle estimation was significantly inferior to that of the proposed model. This suggests that incorporating spatial features can contribute to a more accurate estimation of motor intentions. The proposed model exhibits superior estimation accuracy compared to CNN, despite the latter's capability to extract local spatial information from HD-EMG [27]. This superiority can be attributed to the utilization of GCN, which allows for dynamic graph-based convolutions on HD-EMG. The incorporation of graphical connections and diverse neighbor aggregation methods in GCN empowers the model to capture intricate dependencies and correlations among different EMG channels. This capability proves particularly beneficial in decoding hand movement intention from HD-EMG, where spatial relationships are complex and influenced by synergies [16], [40]. Furthermore, the concatenation approach used to update target vertices' features, akin to residual connections, preserves the original features of target vertices, contributing to a more comprehensive and informative representation.

C. Analysis of the Robustness and Scalability of the Proposed Model

Experimental results indicated that the proposed model maintained acceptable accuracy even with a significant reduction in the number of HD-EMG electrodes. This finding suggests that the extracted spatial features are resilient to variations in the number of electrodes, demonstrating the robustness of the model. This finding also highlights the trade-off between the number of electrodes and the precision of specific control parameters. It has practical implications for prosthesis users aiming to enhance comfort by reducing the number of electrodes on the prosthesis socket while maintaining acceptable control accuracy.

The proposed model exhibits inferior performance in leave-one-subject-out evaluations, highlighting the impact of user-specific variation on the model's robustness. In fact, developing a subject-independent model is pivotal for facilitating the widespread adoption of robust prosthetic hands, but it would also involve overcoming significant challenges related to variability between subjects since the forearm anatomy and motor unit firing mechanisms are all essential factors that can influence individual muscle activation patterns [46]. Future work should address these challenges to navigate the complexities introduced by user-specific variations in muscle activation patterns.

In terms of computational efficiency, execution time is less than 3 ms per frame for the proposed model, as evaluated on a personal computer with a CPU R9-3950X and GPU RTX3070Ti. Despite the light computational burden, the proposed model, requiring data from 6 successive windows for an initial prediction, introduces a 600 ms delay, restricting its real-time applicability in certain practical scenarios. Future work should focus on reducing the model's computational burden and data requirements, facilitating real-time, on-chip processing in prosthetic systems.

D. Limitations and Future Work

There are several limitations to this study. First, the lack of amputee participants in the experiment limits the validation of the proposed model in real clinical scenarios. Future research should involve amputees to assess the performance and usability of the model in practical applications. Second, we achieved SPC only for predefined patterns of wrist movements and grasp force, without considering the impact of upper limb positions on estimation accuracy. Prosthetic users, however, may execute arbitrary wrist movements and variations in grasp force from different upper limb positions. Future efforts should focus on designing algorithms to address the challenges posed by arbitrary movement patterns and upper limb positions. Furthermore, this study is limited to only evaluating the model's performance under ideal laboratory conditions, without assessing the robustness of the model's accuracy to real-life factors such as cross-day variations, electrode degradation, and muscle fatigue. Future research should focus on enhancing the long-term reliability and repeatability of simultaneous and proportional kinematic and kinetic control. Although the proposed model can extract effective spatial-temporal features, it remains a black box model, which hinders its practical application in commercial prostheses. Future research should aim to enhance the interpretability of deep learning models in SPC. Additionally, exploring the integration of physiological information, such as motor unit spiking trains, with deep learning models holds promise for improving overall performance.

V. Conclusion

In this paper, we presented a novel multi-task, spatial-temporal model driven by graphical HD-EMG for simultaneous and proportional wrist angle and grasp force estimation. The results demonstrated the superior performance of the proposed model compared to LSTM, ConvLSTM, CNN, SVR, and GPR. In addition, the study revealed a significant reduction in estimation error for grasp force compared to wrist angle estimation within three multi-DoF movements. In addition, the presented article highlighted the possibility of reducing the number of electrodes from the complete configuration while maintaining acceptable wrist angle and grasp force estimation performances. This study offers a novel approach to achieving SPC in myoelectrical prostheses, potentially augmenting the naturalness and intuitiveness of human-machine interaction.

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